UIT RGPV SHIVPURI

TECHNOVERSE
Future Tech Innovators

AI/ML • Agentic Systems Track

BUILD AN AUTONOMOUS JOB APPLICATION AGENT
AI-Powered Multi-Agent Career Assistant with React Dashboard
BACKGROUND CONTEXT

In today's competitive job market, candidates spend a significant amount of time manually creating resumes, searching for relevant jobs, and applying across multiple platforms.
This process is:

- Time-consuming
- Repetitive
- Inefficient
- Often results in missed opportunities

Candidates also struggle with:

- ATS (Applicant Tracking System) optimization
- Resume keyword matching
- Tracking application status across platforms

CORE CHALLENGE

Build an **AI-powered Autonomous Job Application System** that acts as a **personal job assistant** using a **multi-agent architecture**.
The system must:

- Collect complete user profile data
- Generate ATS-optimized resumes
- Fetch relevant job opportunities from multiple platforms
- Automatically apply (or simulate applying) to jobs
- Track application status in real time
- Display everything in a **production-ready React dashboard**

CORE FEATURES

1. User Profile Input Agent

Collect structured user data:

- Skills
- Experience
- Projects
- Education
- Certifications
- Portfolio / GitHub
- Personal details

2. Resume Agent (ATS Optimizer)

- Generate ATS-friendly resume automatically
- Optimize keywords based on job descriptions
- Provide ATS score
- Suggest improvements

3. Job Finder Agent
Fetch jobs from minimum 4 platforms:

- Naukri.com
- LinkedIn
- Indeed
- Glassdoor

System should:

- Filter jobs based on skills
- Match job descriptions with resume
- Rank opportunities

4. Auto Apply Agent

- Automatically apply to jobs OR simulate application process
- Fill forms / trigger dummy APIs
- Log application activity

5. Notification Agent

- Provide real-time updates:
- “Application submitted successfully”
- “Response received”
- “Interview scheduled”

6. React Dashboard (MANDATORY)
Primary evaluation interface
Must include:
1. Input Section
User profile form
“Generate Resume & Apply” button
Loading indicator
2. Resume Section

- Generated resume preview
- ATS score display
- Improvement suggestions

3. Job Matches Panel

- List of matched jobs
- Platform source
- Match score

4. Applications Tracker
Table:
Job Title | Company | Platform | Status | Timestamp
Status:

- Pending

- Applied
- Response

5. Notifications Panel
Real-time updates
Activity log
TECHNICAL REQUIREMENTS
Frontend
- React / Next.js
- Responsive UI
- Deployed publicly

Backend
REST API
AI Integration
Resume generation (LLM or template-based)
Keyword matching
Job recommendation logic
Architecture
Multi-agent system (LangGraph / CrewAI / AutoGen recommended)
Modular agent design
EXPECTED SYSTEM FLOW
User Input → Resume Generation → Job Matching → Auto Apply → Status Tracking → Dashboard Display
EVALUATION CRITERIA

Criterion	Points	Description
Functionality	30	End-to-end working system
AI Innovation	20	Smart resume + job matching
Dashboard Quality	20	UI/UX + responsiveness
Accuracy	15	Job relevance & ATS scoring
Architecture	10	Multi-agent design
Bonus Features	5	APIs, extensions, notifications